DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103

Semester Examination – May - 2019

Branch: Electronics and Telecommunication **Sem.:- IV**

Subject with Subject Code:- Electrical Machines and Instrumentation (BTESC401) **Marks: 60**

Date:- 14/05/2019 **Time:- 3 Hr.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

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(Marks)

Q.1. **(12)**

a. What is the principle of operation of a dc generator? (02)

b. State with reasons if the following statements are 'true' or 'false: (04)

i. Lap winding is useful for low voltage, high current machines.

ii. Wave winding is useful for high voltage, low current machines.

c. A 4 pole lap wound dc motor has 480 conductors. The flux per pole is 24mWb and the armature resistance is 1Ω. If the motor is connected to a 200 V dc supply and running at 1000 rpm on no load, calculate:

i. Back emf ii. Armature current

iii. Power output iv. Lost torque (06)

Q.2. **(12)**

a. State the difference between transformer and induction motor (three points) (03)

b. State and explain types of synchronous motor. (03)

c. A particular load is to be driven at about 700 r.p.m. What should be the number of poles for a three phase induction motor when : 1. f = 60 Hz 2. f = 50 Hz ? Calculate the actual speed in each case if rated slip is 4%. (06)

Q.3. **(12)**

a. State and explain the types of stepper motor. (03)

b. Give the comparison between AC and DC servomotors. (any 3 points) (03)

c. With the help of the neat diagrams, explain the operation of the capacitor start induction motors. (06)

Q.4. **(12)**

a. Define transducer and explain the classification of transducer. (03)

b. Explain construction and working principle of LVDT. What is meant by residual voltage? (04)

c. A platinum thermometer has a resistance of 100 Ω at 25°C. Find its resistance at 65°C if the platinum has a resistance temperature coefficient of 0.00392 / °C. (05)

Q.5. **(12)**

a. What is accelerometer? Enlist the types of accelerometer. (04)

b. Explain the principle of sound measurement. (04)

c. Write a short note on dynamometer type wattmeter. (04)

Q.6. **(12)**

a. Give classification of recorders. (03)

b. Explain with neat diagram galvanometer type strip chat recorder. (03)

c. Write a short note on: (06)

i. circular chat recorder

ii. X-Y recorder

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103

Semester Examination – Nov - 2019

Branch: Electronics and Telecommunication **Sem.:- IV**
Subject:- Electrical Machines and Instruments (BTESC401) **Marks: 60**
Date:- 26/11/19 **Time:- 3 Hr.**

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly

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			Marks
Q.1			
	a)	Derive the e.m.f. equation of the d.c. machine. State clearly the meaning and units of the symbols used.	6
	b)	A 220 V dc shunt machine has an armature resistance of 0.5 ohm. If the full load armature current is 20A, find the induced emf when the machine acts as: 1. Generator 2. Motor	6
Q.2			
	a)	What are the different methods of speed control of 3-phase induction motor?	6
	b)	Explain the construction of synchronous generator.	6
Q.3			
	a)	Explain the working principle of permanent magnet stepper motor.	6
	b)	Explain the operation of a 2-phase variable reluctance motor.	6
Q.4			
	a)	Explain the principle, working and construction of LVDT. What is meant by residual voltage?	6
	b)	A platinum thermometer has a resistance of 100Ω at 25°c. Find its resistance at 65°c if the platinum has a resistance temperature coefficient of 0.00392/°C.	6
Q.5			
	a)	Explain different methods for measurement of thickness.	6
	b)	Write a short note on: Digital tachometer.	6
Q.6			
	a)	Give classification of recorders.	6
	b)	Explain with neat diagram galvanometer type strip chart recorder.	6

Paper End

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : E &TC Semester :III Subject Code & Name: BTES304 Electrical Machines and Instruments Max Marks: 60 Date: Duration: 3 Hr.		
	Instructions to the Students: *1. All the questions are compulsory.* *2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.* *3. Use of non-programmable scientific calculators is allowed.* *4. Assume suitable data wherever necessary and mention it clearly.*		
		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	With suitable diagram explain construction of DC machine.		**06**
B)	With the help of neat diagram explain three point starter of DC machine.		**06**
C)	List various speed control methods of DC motors. Explain any one.		**06**
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain autotransformer starter for induction motor with the help of suitable diagram.		**06**
B)	What is hunting? Explain causes, effect of hunting in synchronous motors.		**06**
C)	Explain working principle of synchronous motors with suitable diagram.		**06**
Q. 3	**Solve Any Two of the following.**		**12**
A)	Explain working of stepper motor with the help of neat diagram.		**06**
B)	Explain working of Linear induction motor with the help of neat diagram.		**06**
C)	Explain working of FHP motor with the help of neat diagram.		**06**
Q.4	**Solve Any Two of the following.**		**12**
A)	With the help of suitable diagram explain working of LVDT.		**06**
B)	List different temperature transducers. Explain any one.		**06**
C)	With the help of suitable diagram explain working of Hall effect transducer.		**06**
Q. 5	**Solve Any Two of the following.**		**12**
A)	With the help of suitable diagram explain working of sound level meter.		**06**
B)	With the help of block diagram explain working of X-Y plotter.		**06**
C)	Explain working of object counter with the help of neat diagram.		**06**
	***** End *****		

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. S.E **Branch : Electronics Engineering** **Semester : IV**

Subject Code & Name: Electrical Machines and Instruments (BTES401)

Max Marks: 60 **Date: 12/08/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		
A)	A 220V Shunt Motor has Ra = 0.5 Ω, Ia = 40A. Rsh= 220 Ω Find Back EMF, Load Current and Shunt Current	**CO1,CO2, CO3**	**6**
B)	Explain Construction and Working Principe of DC Generator	**CO1,CO2, CO3**	**6**
C)	Derive EMF Equation of DC Machine	**CO1,CO2, CO3**	**6**
Q.2	**Solve Any Two of the following.**		
A)	What is Synchronous Condenser? Explain with the help of Phasor Diagram its Operation	**CO1,CO2, CO3**	**6**
B)	Explain the Procedure of V- Curve in case of Synchronous Motor	**CO1,CO2, CO3**	**6**
C)	Explain Construction and Working Principle of Induction Motor	**CO1,CO2, CO3**	**6**
Q. 3	**Solve Any Two of the following.**		
A)	What is the Principle and Working of Hysterisis Motor? Explain Briefly	**CO1,CO2, CO3**	**6**
B)	Explain the working of Variable Reluctance Stepper Motor	**CO1,CO2, CO3**	**6**
C)	How Many Types of Servomotors, Explain in Details?	**CO1,CO2, CO3**	**6**
Q.4	**Solve Any Two of the following.**		
A)	What is the need of Signal Conditioning and Explain its Types	**CO4**	**6**
B)	How Strain- Gauge works? Explain its Types.	**CO4**	**6**
C)	State selection criteria of Transducer for suitable application	**CO4**	**6**
Q. 5	**Solve Any Two of the following.**		
A)	Explain XY recorder and its applications	**CO4, CO5**	**6**
B)	Explain Electrical Telemetry, Thickness & Humidity measurement	**CO4, CO5**	**6**
C)	How gas analyzer works?	**CO4, CO5**	**6**

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer-2023

Course: B. Tech. **Branch: Electronics Engineering** **Semester: IV**

Subject Code & Name: BTES401& Electrical Machines and Instruments

Max Marks: 60 **Date: 13-07-2023** **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Explain the Construction of a DC Machine with a neat diagram	CO 1	6
B)	Describe any two characteristics of a DC generator.	CO 2	6
C)	Derive the equation of EMF for a DC Machine. State clearly the meaning and units of the symbols used.	CO 1	6
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain the Construction and working principle of a 3-phase induction motor.	CO 1	6
B)	What is hunting? Explain its causes and prevention in synchronous motors.	CO 3	6
C)	What are the main components of a synchronous motor? list the several applications for synchronous motors.	CO 4	6
Q. 3	**Solve Any Two of the following.**		**12**
A)	Explain the working of a servo motor with its principle	CO 2	6
B)	Explain the operation of a three-phase variable reluctance motor.	CO 2	6
C)	Describe the Difference between Stepper Motor and Servo Motor in ten points.	CO 2	6
Q.4	**Solve Any Two of the following.**		**12**
A)	Explain the strain gauge transducer with neat diagrams.	CO 4	6
B)	What is signal conditioning? Explain any one type of signal conditioning.	CO 4	6
C)	Explain the construction and working of the LVDT.	CO 4	6
Q. 5	**Solve Any Two of the following.**		**12**
A)	Define vibration, electrical telemetry, thickness, humidity, and thermal conductivity with one example in terms of electronics.	CO 4	6

B)	Define telemetry. Explain the telemetry system using a neat block diagram.	CO 5	6
C)	Explain the X-Y Recorder and list its applications.	CO 4	6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech. **Branch: Electronics Engineering** **Semester: IV**

Subject Code & Name: Electrical Machines and Instruments (BTES401)

Max Marks: 60 **Date: 16/1/2024** **Duration: 3Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	**Define DC Generator. Describe the Self-excited types of DC generators.**	**CO1**	**6**
B)	**Derive the equation of EMF for a DC Machine. State the meaning and units of the symbols used.**	**CO1**	**6**
C)	**Explain the Construction of a DC Machine with a neat diagram.**	**CO1**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	**What is hunting? Explain its causes and prevention in synchronous motors.**	**CO3**	**6**
B)	**Write a Short note on the synchronous condenser.**	**CO3**	**6**
C)	**Explain the Construction and working of a 3-phase induction motor.**	**CO2**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	**Explain an AC servo motor. List the applications of a servo motor.**	**CO2**	**6**
B)	**Explain the working mechanism with a neat diagram and state the applications of a Hysteresis motor.**	**CO3**	**6**
C)	**Explain the operation of a variable reluctance motor with its principle.**	**CO3**	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	**Explain the operating principle of Hall Effect transducers. Discuss any two applications where these transducers are commonly used.**	**CO4**	**6**
B)	**Explain the strain gauge transducer with its application.**	**CO4**	**6**
C)	**Explain the construction and working of the LVDT.**	**CO4**	**6**

Q. 5	**Solve Any Two of the following.**		**12**
A)	**What is telemetry? Explain the telemetry system using a neat block diagram.**	**CO5**	**6**
B)	**Write a short note on the digital Tachometer.**	**CO5**	**6**
C)	**Define vibration, electrical telemetry, thickness, humidity, and thermal conductivity with one example in terms of electronics.**	**CO5**	**6**

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Summer -2024

Course: B. Tech. **Branch : Electronics Engineering** **Semester :IV**

Subject Code & Name: BTES401 Electrical Machines and Instruments

Max Marks: 60 **Date:** 12/06/2024 **Duration: 3 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Explain the construction and working of DC motor.		**6**
B)	What are the different types of DC generator? Explain anyone with the help of diagram.		**6**
C)	Derive the EMF equation of DC motor and generator.		**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain the construction and working of Induction motor		**6**
B)	Explain the construction and working of Synchronous motor		**6**
C)	Write a short note on synchronous condenser.		**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	Explain the construction and working of Servo motor.		**6**
B)	Explain the construction and working of Stepper motor.		**6**
C)	Write a short note on variable reluctance motor.		**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	Explain the LVDT with the help of diagram.		**6**
B)	Explain the different types strain gauges in brief.		**6**
C)	Write a short note on classification and selection of transducer.		**6**
Q. 5	**Solve Any Two of the following.**		**12**
A)	Explain X-Y recorder with the help of block diagram and enlist its applications.		**6**
B)	Enlist the transducers for humidity measurement and explain anyone in detail.		**6**

C)	Write a short note on burglar alarm and tachometer.		**6**
	***** End *****		